

SHIFAA

Business Requirements Document

Intelligent Patient Journey Management Platform

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Prepared For	SHIFAA Graduation Project	Prepared By	SHIFAA Project Team
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Purpose of this document:

This BRD defines the business context, objectives, scope, stakeholders, functional requirements, non-functional requirements, business rules, data requirements, security expectations, and future direction of the SHIFAA platform. It is the business baseline connecting the project vision to the technical design.

Source basis: the supplied SHIFAA - Document Project Architecture Overview, together with the established project database/ERD design. Source-supported requirements are preserved; implementation details not explicit in the source are marked **Derived / Proposed**.

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1. Document Purpose & Business Context

SHIFAA is an integrated healthcare platform designed to manage the complete Patient Journey, beginning with registration and initial assessment and continuing through diagnosis, treatment, monitoring, recovery, and long-term follow-up. The architecture overview presents it as a Unified Ecosystem connecting patients, doctors, hospitals, family/caregivers, telemedicine services, wearables, and AI capabilities.

1.1 BRD Objectives

- Translate the project vision into a clear and testable requirements baseline.
- Define capabilities for patients, doctors, hospitals, caregivers, and authorized users.
- Align business requirements with the established 31-entity ERD.
- Provide a reference for SRS, UI/UX, API, database, testing, and project management.

1.2 Requirement Classification

Confirmed: directly supported by the supplied architecture overview. **Derived / Proposed:** implementation-level detail derived from the project scope and ERD and requiring validation before being treated as mandatory.

2. Project Overview

2.1 Product Definition

SHIFAA combines electronic medical information, appointment and treatment management, remote care, wearable/device monitoring, AI-assisted healthcare, alerts, family/caregiver access, and operational dashboards.

2.2 Core Platform Components

Component	Business Purpose
Patient Mobile Application	Patient access to medical records, tests, images, vital signs, wearables, reminders, treatment, and AI assistance.
Doctor Dashboard	Clinical workspace for patient history, visits, tests, treatment plans, trends, and alerts.
Hospital Dashboard	Operational and quality monitoring including capacity/utilization, staff performance, adherence, readmission, and healthcare quality.
Family Portal	Controlled access for authorized caregivers/family members and relevant alerts.
Telemedicine Module	Secure chat, audio/video consultations, and exchange of clinical documents/recommendations.
AI Modules	Health assistance, predictive risk analysis, medical summarization, medication adherence analysis, and recovery/anomaly monitoring.

3. Business Problem & Opportunity

3.1 Problems Identified in the Project Overview

Area	Challenge	Business Impact
Data fragmentation	Records are distributed across paper files, clinics, and disconnected imaging/analysis centers.	Loss of medical history and repeated tests.
Post-visit disconnection	Communication may stop after the patient leaves the clinic or hospital.	Acute complications may be detected late.
Treatment adherence	Patients may forget medication or stop treatment without review.	Higher relapse/readmission risk.
Doctor time	Doctors spend substantial time reviewing files and previous reports.	Lower clinical efficiency and higher workload.

3.2 Proposed Business Response

The platform creates a unified digital ecosystem in which health data can be collected, organized, monitored, analyzed, and presented to the appropriate user. AI supports early guidance, risk prediction, summarization, adherence analysis, and anomaly detection rather than replacing clinical judgment.

4. Vision, Objectives & Success Measures

4.1 Vision

Become an end-to-end Patient Journey Platform and a digital healthcare engine integrating electronic medical records, intelligent monitoring, and telemedicine to support proactive, data-driven healthcare.

4.2 Business Objectives

- Centralize relevant patient medical information.
- Improve continuity of care before, during, and after visits.
- Enable faster access to longitudinal information for authorized doctors.
- Support medication adherence through reminders, tracking, and analytics.
- Enable real-time and remote communication.
- Support early risk detection and proactive healthcare workflows.
- Give caregivers controlled visibility where authorization permits.
- Provide hospitals with operational and quality analytics.
- Maintain strong security, access control, auditability, and data protection.

4.3 Suggested Success Measures (Derived / Proposed)

KPI	Direction
Appointment completion rate	Increase
Medication adherence rate	Increase
Readmission rate	Decrease
Time to review patient history	Decrease
Time to detect critical anomaly	Decrease

KPI	Direction
Patient engagement with reminders	Increase
Unauthorized-access incidents	Minimize / Zero tolerance

5. Stakeholders & User Roles

Stakeholder	Primary Need	Interaction
Patient	Manage health journey, records, treatment, monitoring, communication.	Mobile App
Doctor	Review history; manage diagnosis, tests, treatment, and risk.	Doctor Dashboard
Caregiver / Family	Support an authorized patient and receive relevant alerts.	Family Portal
Hospital / Management	Monitor operational and quality indicators.	Hospital Dashboard
Administrator (Derived/ Proposed)	Manage users, configuration, access, and operational controls.	Admin Functions
AI Services	Analyze supported data and generate AI outputs.	AI Modules

5.1 Access Principle

Access must be based on role and explicit patient permissions where required. The architecture overview specifically identifies RBAC and explicit patient consent before sharing patient data.

6. Scope

6.1 In Scope

- User accounts and roles.
- Patient, doctor, caregiver, hospital, and department information.
- Patient-caregiver authorization.
- Appointments and visits.
- Assessments and diagnoses.
- Medical documents, tests, results, and images.
- Vital signs, wearable/device data.
- Treatment plans, prescriptions, medicines, and medication tracking.
- Alerts and notifications.
- Telemedicine online visits, chats, participants, and messages.
- Consent and audit logging.
- Five identified AI modules.
- Doctor and hospital dashboards with longitudinal/time-series analytics.

6.2 Future / Expansion Scope

- External EHR/HIS, pharmacy, insurance, and hospital-system integration.
- Advanced Medical Imaging Analysis.
- Broader IoT health-device connectivity.
- Production regulatory certification and clinical validation unless separately approved.

7. Patient Journey

Stage	Business Flow	Key Capabilities
1. Registration & Setup	Create account, upload files and medical/insurance information.	User, patient profile, documents, consent.
2. Initial Assessment	AI interaction to identify symptoms and risk level.	AI Health Assistant, assessment, risk indication.
3. Appointment & Consultation	Book appointment or initiate remote consultation.	Appointment, doctor interaction, telemedicine.
4. Examination & Diagnosis	Doctor reviews file, records diagnosis, and documents treatment.	Visit, assessment, diagnosis, tests, images, treatment.
5. Monitoring & Follow-up	Treatment reminders plus vital-sign/device monitoring.	Medication tracking, vital signs, devices, notifications.
6. Early Warning & Recovery	Analyze data, detect risk, issue alerts, and follow recovery.	AI risk analysis, anomaly detection, alerts, longitudinal analytics.

8. Functional Requirements

8.1 Identity, Accounts & Profiles

ID	Requirement	Priority	Basis
FR-001	Users shall be able to create and maintain an account using a unique email and secure password representation.	Must	Derived / ERD
FR-002	Each applicable account shall have a role such as Patient, Doctor, Caregiver, or administrative role.	Must	Project + ERD
FR-003	Patient, Doctor, and Caregiver profile data shall be maintained separately from credentials.	Must	ERD
FR-004	The system shall maintain account status and relevant login timestamps.	Should	ERD

8.2 Patient & Caregiver Management

ID	Requirement	Priority	Basis
FR-005	The system shall maintain patient medical record number, demographics, blood type, contact/emergency information, and status.	Must	ERD
FR-006	A patient shall be able to authorize a caregiver through a relationship with permission level and validity period.	Must	Project + ERD
FR-007	Caregiver access shall follow active authorization and permission level.	Must	Project + Security

8.3 Appointments & Clinical Visits

ID	Requirement	Priority	Basis
FR-008	Patients shall be able to book appointments with doctors and maintain date, time, type, reason, notes, and status.	Must	Project + ERD

ID	Requirement	Priority	Basis
FR-009	A clinical visit shall be recorded from an appointment when a consultation occurs.	Must	ERD
FR-010	The system shall support assessments and diagnoses associated with a visit.	Must	Project + ERD

8.4 Medical Records, Tests & Imaging

ID	Requirement	Priority	Basis
FR-011	Authorized users shall be able to store and retrieve patient medical documents.	Must	Project + ERD
FR-012	The system shall record document uploader and upload date.	Must	ERD
FR-013	The system shall support medical tests requested during visits and their status/notes.	Must	ERD
FR-014	The system shall store test result value, unit, normal range, interpretation, and notes where applicable.	Must	ERD
FR-015	The system shall support X-Ray, CT, MRI, and Ultrasound images with reports.	Must	Project + ERD

8.5 Treatment & Medication

ID	Requirement	Priority	Basis
FR-016	Doctors shall be able to create treatment plans and optionally associate them with a diagnosis.	Must	Project + ERD
FR-017	Prescriptions shall be linked to treatment plans.	Must	ERD
FR-018	Prescriptions shall support multiple medicines through prescription items.	Must	ERD
FR-019	Medication events shall track scheduled time, taken time, status, and notes.	Must	Project + ERD
FR-020	The system shall provide medication reminders and adherence-related notifications.	Must	Project

8.6 Monitoring, Alerts & Notifications

ID	Requirement	Priority	Basis
FR-021	Patients shall be able to connect supported health devices and maintain connection metadata/status.	Must	Project + ERD
FR-022	Timestamped device readings shall be stored with type, value, and unit.	Must	Project + ERD
FR-023	The system shall support patient vital-sign records independent of device readings when necessary.	Must	Project + ERD
FR-024	The system shall create alerts for relevant risk, abnormality, adherence, or emergency conditions.	Must	Project + ERD
FR-025	Alerts shall track severity, status, creation time, and resolution time.	Must	ERD

ID	Requirement	Priority	Basis
FR-026	Notifications shall be delivered to authorized users for reminders, alerts, appointments, and supported events.	Must	Project + ERD
FR-027	Notifications shall track read state and status.	Must	ERD

9. AI Modules Requirements

The architecture overview explicitly identifies five AI capabilities. These requirements describe business behavior without prescribing a specific ML algorithm.

AI Module	Business Capability	Expected Value
AI Health Assistant	Collect symptoms, medication information, measurements, and initial context; provide early guidance and specialty direction.	Earlier guidance without presenting final medical diagnosis.
Predictive Risk Analysis	Analyze cumulative health data to predict potential complications before they occur.	Shift toward proactive/preventive care.
AI Medical Summary	Summarize complex and longitudinal medical files for doctors.	Reduce review time.
Medication Adherence Analysis	Track medication behavior and detect risk of interruption/non-adherence.	Enable early awareness and intervention.
Recovery Monitoring & Anomaly Detection	Monitor recovery trajectory and identify sudden/abnormal vital-indicator changes.	Support rapid clinical attention.

9.1 AI Safety & Human Oversight (Derived / Proposed)

- AI outputs should be clearly distinguished from clinician-authored diagnoses and treatment decisions.
- High-risk alerts should be routed to an appropriate authorized care provider.
- Where feasible, retain the context/data window contributing to an AI-supported alert or summary.
- AI confidence, limitations, and escalation rules should be defined before production use.

10. Telemedicine Requirements

ID	Requirement	Priority
TR-001	Online visits shall be linked to appointments.	Must
TR-002	The system shall support real-time text chat between authorized participants.	Must
TR-003	The system shall support audio/video consultation capability.	Must
TR-004	The system should allow exchange of medical documents and recommendations during consultation.	Should
TR-005	Online visits shall maintain start/end times, type, status, meeting reference, and notes where applicable.	Must
TR-006	Telemedicine access shall respect authentication, authorization, and patient consent.	Must

11. Data & Information Requirements

11.1 Data Domains

Domain	Representative Data
Identity & Access	User, role, status, login metadata.
Patient Profile	Demographics, medical record number, blood type, contact/emergency information.
Care Relationships	Patient-caregiver permissions and validity.
Clinical Care	Appointments, visits, assessments, diagnoses.
Clinical Evidence	Medical documents, tests, results, images.
Monitoring	Vital signs, devices, device readings.
Treatment	Treatment plans, prescriptions, medicines, prescription items, medication tracking.
Communication	Online visits, chats, participants, messages.
Engagement & Safety	Alerts, notifications.
Privacy & Accountability	Consent, audit logs.

11.2 Data Quality (Derived / Proposed)

- Patient identifiers should be unique and consistently referenced.
- Date/time values should preserve the timing needed for medication and device monitoring.
- Documents and images should retain source/upload metadata.
- Status, role, type, severity, and similar controlled values should use consistent allowed values.
- Historical clinical records should not be silently overwritten where auditability is required.

12. Security, Privacy & Compliance

ID	Requirement	Priority	Source
SEC-001	Sensitive health data shall be protected using encryption in transit and at rest.	Must	Project overview
SEC-002	The system shall enforce Role-Based Access Control (RBAC).	Must	Project overview
SEC-003	Explicit patient consent shall be required before sharing protected patient data where applicable.	Must	Project overview
SEC-004	The system shall maintain comprehensive audit logs for security/accountability.	Must	Project overview
SEC-005	Multi-factor authentication (MFA) should be supported for protected accounts and sensitive workflows.	Should	Project overview
SEC-006	Patient information access shall follow least-privilege principles.	Must	Derived / Proposed

13. Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-001	Security	Sensitive data shall be encrypted in transit and at rest.	Must

ID	Category	Requirement	Priority
NFR-002	Availability	Core patient and clinical workflows should be reliably available during supported operating periods.	Must
NFR-003	Performance	Common dashboard and record retrieval should provide an acceptable user-facing response time under expected load.	Should
NFR-004	Scalability	Architecture should support growth in patients, records, readings, messages, and AI workloads.	Should
NFR-005	Usability	Patient workflows should be simple for routine appointments, records, reminders, and alerts.	Must
NFR-006	Auditability	Security-sensitive actions and relevant data changes shall be traceable.	Must
NFR-007	Maintainability	Business rules and data structures should be modular for future integrations and AI modules.	Should
NFR-008	Interoperability	Solution should be designed for future EHR/HIS, pharmacy, insurance, and IoT integration.	Should

14. Business Rules

Rule	Business Rule
BR-001	Every Patient, Doctor, and Caregiver profile must be associated with a valid User account.
BR-002	A User may exist without one of these profiles when the role does not require a healthcare profile.
BR-003	Caregiver access is permitted only while applicable authorization/consent is active.
BR-004	A Visit is associated with an Appointment in the current model.
BR-005	Assessments, diagnoses, tests, and images are recorded in visit context.
BR-006	A TreatmentPlan belongs to a patient and is created by a doctor; diagnosis association is optional.
BR-007	A Prescription belongs to a TreatmentPlan and contains PrescriptionItems.
BR-008	Medication Tracking records medication events against a prescription item and patient.
BR-009	DeviceReadings belong to a Device and inherit patient context through the device.
BR-010	A Notification may originate from an Alert, but not every Notification must be an Alert.
BR-011	Online Visits are linked to Appointments.
BR-012	Messages belong to a Chat and are sent by authenticated users.
BR-013	Audit logs retain acting user, action, affected entity, timestamp, and relevant technical context.

15. Integrations & External Interfaces

Interface	Purpose	Status
Wearables / Health Devices	Receive health measurements for monitoring.	Core
Telemedicine Services	Secure audio/video and real-time communication.	Core
File / Medical Image Storage	Store medical documents/images referenced by records.	Core

Interface	Purpose	Status
Notification Channels	Deliver reminders and alerts.	Core

15.2 Future Integrations

- Hospital EHR / HIS systems.
- Pharmacy systems.
- Insurance companies.
- Advanced medical imaging analysis services.
- Broader IoT health-device ecosystem.

16. Reporting & Analytics

16.1 Doctor Dashboard

- Unified patient record view.
- Longitudinal/time-series comparison of vital measurements and clinical indicators.
- Treatment and medication overview.
- Alerts and abnormal-pattern visibility.
- AI-generated medical summary and risk-support information.

16.2 Hospital Dashboard

- Operational/capacity indicators.
- Doctor and staff performance indicators.
- Recovery/adherence monitoring.
- Readmission rates.
- Healthcare quality analytics.

17. Future Roadmap

Roadmap Item	Description	Expected Benefit
EHR / HIS Integration	Integrate hospital information systems and electronic health records.	Unified ecosystem; less duplicate entry.
Medical Imaging Analysis	Expand AI to read/analyze medical images.	Additional decision-support capability.
IoT Health Devices	Connect a broader range of health/medical IoT devices.	Richer continuous monitoring.

18. Assumptions, Dependencies & Constraints

18.1 Assumptions

- Users have compatible devices and internet access for mobile/telemedicine functions.
- Authorized doctors/caregivers are correctly identified.
- AI models receive sufficiently structured and valid inputs.
- Supported device types and communication mechanisms can be defined.
- Medical files/images are stored in an appropriately secured accessible location.

18.2 Dependencies

- Authentication and identity management.
- Secure file/image storage.
- Telemedicine communication infrastructure.
- Notification delivery.
- Supported wearable/device interfaces.
- AI/ML services and data pipelines.

18.3 Constraints / Items Requiring Validation

- The supplied source is a scanned 4-page overview and does not provide a complete field-level requirements specification.
- Exact regulatory/compliance jurisdiction is not defined in the source.
- Exact performance targets, availability SLAs, retention periods, and infrastructure limits are not defined.
- The source does not state whether every AI output must be persisted as a separate database object.
- Hospital/department relationships to doctors and appointments require confirmation if that workflow is required.

19. Acceptance Criteria

ID	Acceptance Criterion
AC-001	A patient can access an account and maintain a patient profile.
AC-002	A patient can book an appointment and the visit can be recorded.
AC-003	Authorized users can access relevant medical information according to role and consent.
AC-004	Doctors can record assessments, diagnoses, tests, results, images, and treatment plans.
AC-005	Prescriptions and medication tracking can drive reminders/notifications.
AC-006	Supported devices can provide timestamped readings linked to the correct patient.
AC-007	Alerts can be created, displayed, resolved, and routed to authorized recipients.
AC-008	Telemedicine workflows can link online consultations to appointments and authenticated participants.
AC-009	AI modules can return defined outputs without presenting them as definitive clinician diagnoses.
AC-010	RBAC, consent controls, encryption, MFA support, and audit logging are implemented or specified for the approved scope.

20. Requirements Traceability

Capability	Primary ERD Entities
Identity & profiles	User, Patient, Doctor, Caregiver
Hospital structure	Hospital, Department
Caregiver access	PatientCaregiver, Consent
Appointments & visits	Appointment, Visit, Assessment, Diagnosis
Medical records	Medical Document, MedicalTest, TestResult, MedicalImage
Monitoring	VitalSign, Device, DeviceReading

Capability	Primary ERD Entities
Treatment & medication	TreatmentPlan, Prescription, Medicine, PrescriptionItem, Medicine Tracking
Alerts & notifications	Alert, Notification
Telemedicine	Online Visit, Chat, ChatParticipant, Message
Accountability	AuditLog

20.1 AI Traceability Note

The current 31-entity ERD represents operational data, while the five AI modules are treated primarily as application/service capabilities. If the implementation must persist model outputs, explanations, model versions, prompts/interactions, prediction scores, or recommendation histories, dedicated AI persistence entities should be added after requirement confirmation.

21. Appendix - ERD / Data Model Alignment

21.1 Established 31-Entity Baseline

#	Entity	#	Entity
1	User	17	TreatmentPlan
2	Patient	18	Prescription
3	Doctor	19	Medicine
4	Caregiver	20	PrescriptionItem
5	Hospital	21	Medicine Tracking
6	Department	22	Device
7	Patient Caregiver	23	DeviceReading
8	Appointment	24	Alert
9	Visit	25	Notification
10	Assessment	26	OnlineVisit
11	Diagnosis	27	Chat
12	Medical Document	28	Chat Participant
13	MedicalTest	29	Message
14	TestResult	30	Consent
15	MedicalImage	31	AuditLog
16	Vital Sign		

21.2 Key Data-Model Notes

- **User** is an account entity; personal profile information is held in Patient, Doctor, and Caregiver.
- **PatientCaregiver** resolves the patient-caregiver relationship and stores permission metadata.
- **ChatParticipant** supports multi-user chat membership.
- **DeviceReading** obtains patient context through Device → Patient.

- **Medical Document.UploadedBy** and **Message.SenderID** reference User because these actions are performed by authenticated accounts.
 - **Consent** uses GrantedToUserID so access can be granted to an authorized user without role-specific consent tables.
 - All 31 entities currently have their own primary key; therefore associative entities are not strict textbook weak entities in classic Chen notation.
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End of Document

This BRD is the proposed business baseline for SHIFAA. Changes to scope, workflows, AI capabilities, security controls, or the data model should be reviewed against this document before implementation changes are accepted.